Technical Document #425: Data Engineering - Comprehensive Overview

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Data lineage tracks the origin and movement of data through systems. It is essential for debugging, compliance, and impact analysis.

ETL (Extract, Transform, Load) is a process that extracts data from source systems, transforms it to fit business rules, and loads it into a target data store.

Stream processing handles continuous data streams in real-time. Apache Kafka and Apache Flink are popular stream processing frameworks.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

Quantum computing promises to solve problems that are intractable for classical computers. It has potential applications in cryptography, drug discovery, and optimization.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Key Takeaways:

- Data lakes store raw data in its native format.
- Stream processing handles continuous data streams in real-time.
- Data warehouses store structured data optimized for analytical queries.